CBSE Class 10 Science Question Paper 2022
Set-1(31/1/1) Solutions

Section A

1. (a) Write the molecular formula of the following carbon compounds:
(i) Methane
(ii) Propane
Ans: (i) Methane: CH_4
(ii) Propane: C_3H_8
(b) Carbon compounds have low melting and boiling points. Why? **2**
Ans: Carbon compounds have low melting and boiling points because they are held together by weak van der Waals forces or London dispersion forces, which require less energy to break.

2. The electrons in the atoms of two elements X and Y are distributed in three shells having 1 and 7 electrons respectively in their outermost shells.
(a) Write the group numbers of these elements in the Modern Periodic Table.
Ans:
- Element X (1 electron in the outer shell): Group 1
- Element Y (7 electrons in the outer shell): Group 17

(b) Write the molecular formula of the compound formed when X and Y combine with each other.
Ans: The compound formed would be XY (e.g., NaCl where X is sodium and Y is chlorine).
(c) Which of the two is electropositive? **2**
Ans: Element X is electropositive.

3. (a) Which of the following flowers will have a higher possibility of self-pollination? Mustard, Papaya, Watermelon, Hibiscus
Ans: Hibiscus

(b) List the two reproductive parts of a bisexual flower. **2**

Ans: Stamen (male reproductive part) and Pistil/Carpel (female reproductive part).

4. Which one of the two multicellular organisms — Spirogyra and Planaria reproduces by regeneration and why? Give an example of any other organism which can also reproduce by the same process. **2**

Ans: Planaria reproduces by regeneration because it can grow back its body parts when cut. Another example is Hydra.

5. (a) What is variation? List two main reasons that may lead to variation in a population. **2**

Ans: Variation is the difference in physical traits of an individual from the group to which it belongs. Two main reasons are genetic mutation and sexual reproduction.

OR

(b) (i) In a cross between violet-flowered plants and white-flowered plants, state the characteristics of the plants obtained in the F_1 progeny.

Ans: The F_1 progeny would all have violet flowers due to the dominance of the violet allele.

(ii) If the plants of F_1 progeny are self-pollinated, then what would be observed in the plants of F_2 progeny?

Ans: The F_2 progeny would show a 3:1 ratio of violet to white flowers.

(iii) If 100 plants are produced in F_2 progeny, then how many plants will show the recessive trait? **2**

Ans: 25 plants will show the recessive trait (white flowers).

6. (a) (i) Name and state the rule to determine the direction of force experienced by a current-carrying straight conductor placed in a uniform magnetic field which is perpendicular to it.

Ans: Fleming's Left-Hand Rule: When the first finger points in the direction of the magnetic field and the second finger in the direction of the current, the thumb will point in the direction of the force.

(ii) An alpha particle while passing through a magnetic field gets projected towards the north. In which direction will an electron project when it passes through the same magnetic field? **2**

Ans: The electron will project towards the south.

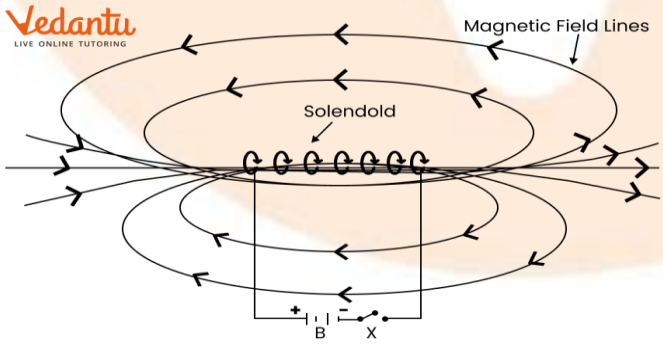
OR

(b) (i) What is a solenoid?

Ans: A solenoid is a coil of wire that generates a magnetic field when an electric current passes through it.

(ii) Draw the pattern of magnetic field lines of the magnetic field produced by a solenoid through which a steady current flows. **2**

Ans:



7. (a) What is ozone? How is it formed in the upper layers of the Earth's atmosphere? How does ozone affect our ecosystem? 2

Ans: Ozone (O_3) is a molecule composed of three oxygen atoms. It is formed in the stratosphere when ultraviolet (UV) radiation from the sun splits an oxygen molecule (O_2) into two oxygen atoms, and one of these atoms combines with another O_2 molecule to form ozone. Ozone protects living organisms by absorbing harmful UV radiation.

OR

(b) (i) List two human-made ecosystems. 1

Ans: Aquariums and Agricultural fields

(ii) "We do not clean a pond in the same manner as we do in an aquarium." Give a reason to justify this statement. 1

Ans: A pond is a natural ecosystem with its own self-regulating processes, whereas an aquarium is a controlled environment that requires manual cleaning and maintenance to keep it healthy.

Section B

8. (a) List two advantages of adopting the atomic number of an element as the basis of classification of elements in the Modern Periodic Table. 3

Ans:

- Consistent Order: Using the atomic number ensures elements are arranged in a consistent and logical order, reflecting their increasing number of protons. This avoids inconsistencies seen in older systems that used atomic mass.
- Predictable Properties: The atomic number arrangement leads to a periodic pattern in element properties. Elements with similar properties fall into the same groups, making it easier to predict the behaviour of unknown or newly discovered elements.

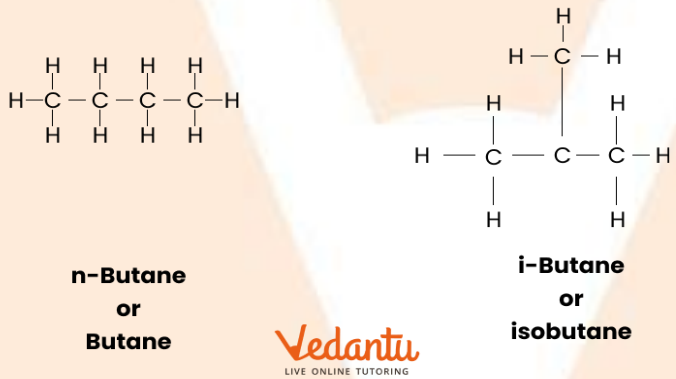
(b) Write the electronic configurations of the elements X (atomic number 13) and Y (atomic number 20). **3**

Ans:

- X (13): 2, 8, 3
- Y (20): 2, 8, 8, 2

9. (a) Draw two different possible structures of a saturated hydrocarbon having four carbon atoms in its molecule. What are these two structures of the hydrocarbon having the same molecular formula called? Write the molecular formula and the common name of this compound. Also, write the molecular formula of its alkyne. **3**

Ans:



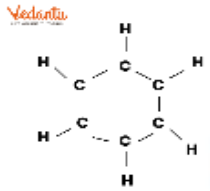
These two structures are called isomers. The molecular formula is C_4H_{10}. Common names are n-butane and isobutane. The alkyne is C_4H_6.

OR

(b) (i) Write the molecular formula of benzene and draw its structure.

Ans: Molecular formula: C_6H_6

Structure:



(ii) Write the number of single and double covalent bonds present in a molecule of benzene.

Ans: Three double bonds and three single bonds

(iii) Which compounds are called alkynes? **3**

Ans: Compounds with a carbon-carbon triple bond are called alkynes.

10. (a) Mention one function of each of the following organs in the human male reproductive system:

(i) Testis

(ii) Scrotum

(iii) Vas deferens

(iv) Prostate gland

Ans:

(i) Testis: The testis is responsible for producing sperm, which is necessary for reproduction. It also produces the hormone testosterone, which is important for male sexual development and characteristics.

(ii) Scrotum: The scrotum is a skin-covered sac that holds and protects the testes. It regulates their temperature by contracting to move the testes closer to the body when it's cold and relaxing to move them away when it's warm.

(iii) Vas deferens: The vas deferens is a long tube that transports sperm from the testes to the urethra during ejaculation. This tube connects the epididymis to the ejaculatory ducts, allowing sperm to be mixed with seminal fluid.

(iv) Prostate gland: The prostate gland produces a fluid that is a key component of semen, helping to nourish and transport sperm. This fluid also helps to protect the sperm and improve its motility.

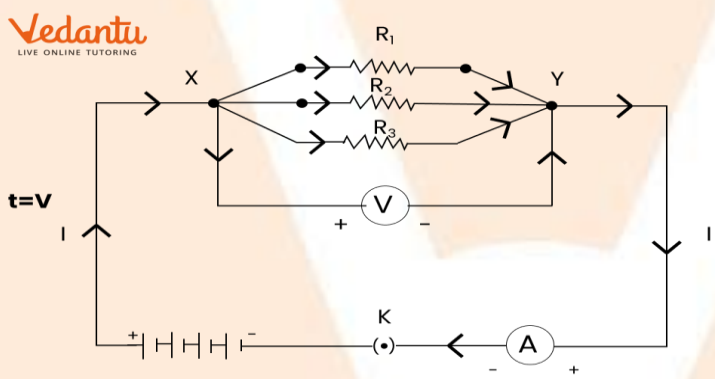
(b) Name the type of germ cell which (i) is motile, and (ii) stores food. **3**

Ans: (i) Motile: Sperm cell

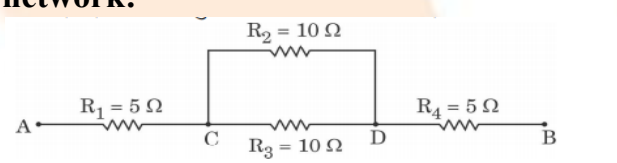
(ii) Stores food: Ovum

11. (a) Three resistors R_1, R_2, and R_3 are connected in parallel and the combination is connected to a battery, an ammeter, a voltmeter, and a key. Draw a suitable circuit diagram to show the arrangement of these circuit components along with the direction of current flowing.

Ans:



(b) Calculate the equivalent resistance of the following network: **3**



Ans:

- For resistors R_2 (10 Ω) and R_3 (10 Ω) in parallel:

$$\frac{1}{R_{eq1}} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \frac{1}{5} \implies R_{eq1} = 5\ \Omega$$

- Then, R_1 (5 Ω) in series with the equivalent resistance:

$$R_{total} = R_1 + R_{eq1} = 5\ \Omega + 5\ \Omega = 10\ \Omega$$

- Finally, this 10 Ω in parallel with R_4 (5 Ω):

$$\frac{1}{R_{final}} = \frac{1}{10} + \frac{1}{5} = \frac{1}{10} + \frac{2}{10} = \frac{3}{10} \implies R_{final} = \frac{10}{3} \approx 3.33\ \Omega$$

12. (a) (i) Define Electric Power and write its SI unit ½ + ½

Ans: Electric power is the rate at which electrical energy is transferred by an electric circuit. The SI unit of electric power is the watt (W).

(ii) Two bulbs rated 100W; 220 V and 60W; 220 V are connected in parallel to an electric main of 220 V. Find the current drawn by the bulbs from the mains **2**

Ans:

For the 100W bulb:

$$P = V \times I$$

$$I = \frac{P}{V} = \frac{100W}{220V} \approx 0.4545A$$

For the 60W bulb:

$$P = V \times I$$

$$I = \frac{P}{V} = \frac{60W}{220V} \approx 0.2727A$$

Total current drawn from the mains:

$$I_{total} = I_{100W} + I_{60W}$$

$$I_{total} \approx 0.4545A + 0.2727A \approx 0.7272A$$

OR

(i) State Joule's law of heating. Express it mathematically when an appliance of resistance R is connected to a source of voltage V and the current I flows through the appliance for a time t.

Ans: Joule's law of heating states that the heat produced in a conductor is directly proportional to the square of the current, the resistance of the conductor, and the time for which the current flows. Mathematically, it is expressed as:

H=I2RtH = I^2RtH=I2Rt

Ans:

Using Joule's law of heating:

$$H = I^2Rt$$

First, find the current I using Ohm's law:

$$V = IR$$

$$I = \frac{V}{R} = \frac{6V}{5\Omega} = 1.2A$$

Then, calculate the heat dissipated:

$$H = (1.2A)^2 \times 5\Omega \times 10s$$

$$H = 1.44 \times 5 \times 10$$

$$H = 72J$$

(ii) A 5 Ω resistor is connected across a battery of 6 volts. Calculate the energy that dissipates as heat in 10 s. 3

Using Joule's law of heating: H=I2RtH = I^2RtH=I2Rt
First, find the current III using Ohm's law: V=IRV = IRV=IR I=VR=6V5Ω=1.2AI = $\frac{V}{R} = \frac{6V}{5O}$ = 1.2AI=RV=5Ω6V=1.2A
Then, calculate the heat dissipated: H=(1.2A)2×5Ω×10sH = =(1.2A)2×5Ω×10s H=1.44×5×10H =1.44×5×10 H=72JH = 72JH=72J

13. (a) Name the group of organisms which form in the first trophic level of all food chains. Why are they called so?

Ans: The group of organisms that form the first trophic level of all food chains are producers, typically plants and algae. They are called producers because they produce their own food through the process of photosynthesis, using sunlight, carbon dioxide, and water.

(b) Why are the human beings most adversely affected by biomagnification?

Ans: Human beings are most adversely affected by biomagnification because they are at the top of the food chain. As toxic substances like pesticides and heavy metals accumulate in the tissues of organisms at each trophic level, the concentration of

these toxins increases. By the time it reaches humans, the concentration is significantly higher, leading to severe health risks and diseases.

© State one ill-effect of the absence of decomposers from a natural ecosystem.
Ans: One ill-effect of the absence of decomposers from a natural ecosystem is the accumulation of dead organic matter, which can lead to nutrient depletion in the soil. Decomposers break down dead plants and animals, recycling nutrients back into the ecosystem, which is essential for the growth of new plants.

SECTION C

This section has 2 case-based questions (14 and 15). Each case is followed by 3 sub-questions (a), (b) and (c). Parts (a) and (b) are compulsory. However, an internal choice has been provided in Part (c).

14. The mechanism by which the sex of an individual is determined is called sex determination. In human beings, the sex of a newborn is genetically determined, whereas in some others it is not. There are 46 (23 pairs) chromosomes in human beings. Out of these, 44 (22 pairs) control the body characters and 2 (one pair) are known as sex chromosomes. The sex chromosomes are of two types X chromosome and Y chromosome. At the time of fertilisation, depending upon which type of male gamete fuses with the female gamete, the sex of the newborn child is decided.

(a) Why is a pair of sex chromosomes in human beings called a mismatched pair in terms of type and size? **½ + ½**

Ans: A pair of sex chromosomes in human beings is called a mismatched pair because the X and Y chromosomes differ in size and genetic content. The X chromosome is larger and carries more genes, while the Y chromosome is smaller and carries fewer genes.

(b) Out of male or female, which of them has a perfect pair of sex chromosomes? In case of a perfect pair, will the gametes produced be of the same kind or of a different kind?

1

Ans: Females have a perfect pair of sex chromosomes (XX). In the case of a perfect pair, the gametes produced will be of the same kind, each carrying one X chromosome. Males, on the other hand, have an XY pair, resulting in gametes that are of different kinds (one carrying an X chromosome and one carrying a Y chromosome).

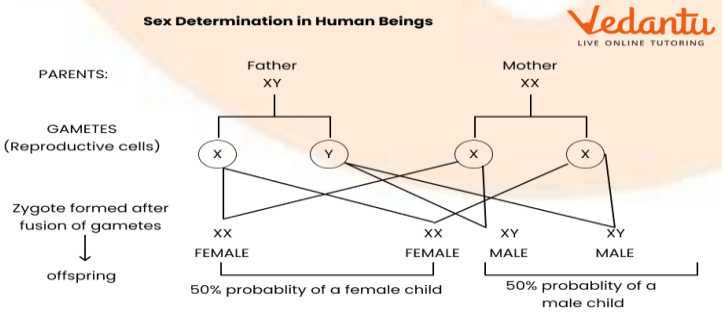
(c) (i) Name two animals whose sex is not genetically determined. Explain the process of their sex determination. 1+1

Ans: Two animals whose sex is not genetically determined are the American alligator and the common snapping turtle.

- American Alligator: In American alligators, sex determination is temperature dependent. The temperature at which the eggs are incubated determines the sex of the offspring. Eggs incubated at lower temperatures (around 30°C) tend to hatch as females, while eggs incubated at higher temperatures (around 34°C) tend to hatch as males.
- Common Snapping Turtle: Similar to alligators, the sex of common snapping turtles is determined by the incubation temperature of the eggs. Eggs incubated at lower temperatures (22-27°C) tend to develop into males, while those incubated at higher temperatures (30°C and above) tend to develop into females.

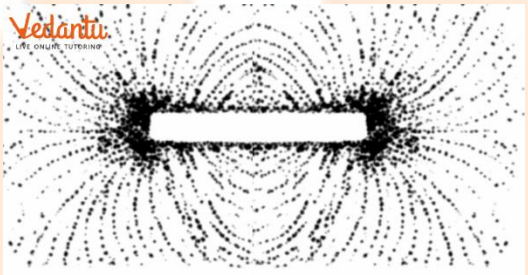
OR

(ii) With the help of a flowchart only, show how sex is genetically determined in human beings.

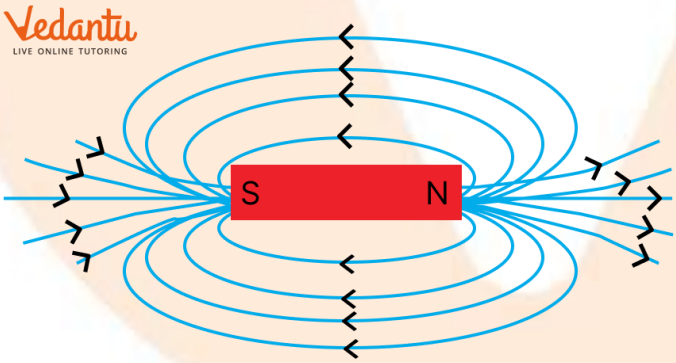


15. A student fixes a sheet of white paper on a drawing board using some adhesive materials. She places a bar magnet in the centre of it and sprinkles some iron filings uniformly around the bar magnet using a salt-sprinkler. On tapping the board gently, she observes that the iron filings have arranged themselves in a particular pattern.

(a) Draw a diagram to show this pattern of iron filings. 1



(b) Draw the magnetic field lines of a bar magnet showing the poles of the bar magnet as well as the direction of the magnetic field lines. 1



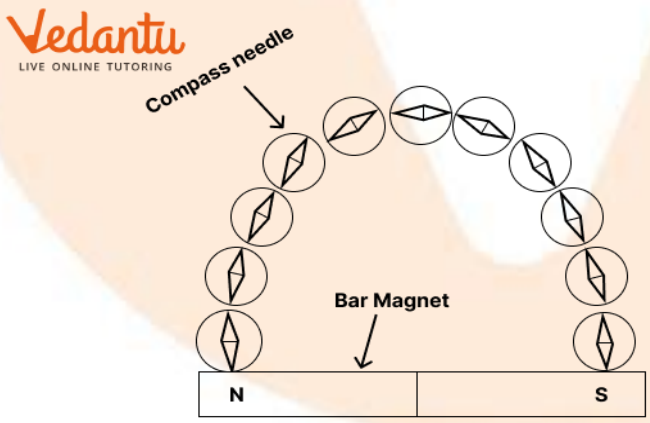
(c) (i) How is the direction of magnetic field at a point determined using the field lines? Why do two magnetic field lines not cross each other? 2

Ans: The direction of the magnetic field at a point is determined by the tangent to the magnetic field line at that point. The direction of the magnetic field is the direction in which a north pole would move if placed at that point.

Two magnetic field lines do not cross each other because at any given point, the magnetic field has a unique direction. If two field lines were to cross, it would imply that the magnetic field at that point has two different directions, which is not possible.

OR

(ii) How are the magnetic field lines of a bar magnet drawn using a small compass needle? Draw one magnetic field line each on both sides of the magnet.

Ans:

- Place the bar magnet in the center of a sheet of paper.
- Use a small compass needle to determine the direction of the magnetic field at various points around the magnet.
- Start at the north pole of the magnet and move the compass in small steps, marking the direction of the needle at each point.
- Connect the marked points to draw a smooth curve from the north pole to the south pole of the magnet.
- Repeat the process for different starting points around the magnet to draw multiple field lines.